WANXING WU M.Sc.

wuwanxing3574@gmail.com
GitHub LinkedIn Homepage

ABOUT

I am an M.S. student in Computer Science at **Institut Polytechnique de Paris (IP Paris)**, with a B.S. in Data Science and Big Data Technology from **Southern University of Science and Technology (SUSTech)**. I am currently a Research Assistant at **SUSTech-NLP**, advised by Prof. Guanhua Chen.

My research interests broadly lie in **efficient LLM systems** and **agentic AI**. I have worked on **LLM–SLM model collaboration** for efficient routing and inference, and I am currently working on **efficient agentic exploration** for test-time adaptation. I am also interested in **LLM for health**, as an application domain where low-resource constraints and trustworthiness are both critical.

EDUCATION

Institut Polytechnique de Paris — Paris, France
M.S. in Computer Science — *Sep 2025 – Jul 2027*

Southern University of Science and Technology (SUSTech) — Shenzhen, China
B.S. in Data Science and Big Data Technology — *Sep 2021 – Jul 2025*

RESEARCH

SUSTech-NLP Group — Shenzhen, China
Research Assistant, advised by Prof. Guanhua Chen. — *Jan 2025 – Present*

- **LLM–SLM Collaboration**: Showed that LLM routers fail silently under distribution shift; built RouterXBench to systematically expose this gap and proposed Dirichlet-based hidden-state probing with distribution-aware layer aggregation for robust routing.

INTERNSHIP

Institute of Intelligent Medical Research (IIMR), BGI — Shenzhen, China
Machine Learning Intern (AI for Genomics). — *Jul 2024 – Jan 2025*

- Developed cfDNA-based feature representations via statistical filtering and model-guided selection; applied ensemble learning for early colorectal cancer detection, achieving **94.88% sensitivity**.

Peking University Shenzhen Hospital — Shenzhen, China
Data Analyst Intern. — *Jul 2023 – Oct 2023*

- Performed biostatistical analysis on multi-timepoint clinical records to compare postoperative outcomes across surgical methods.

PUBLICATIONS

Towards Fair and Comprehensive Evaluation of Routers in Collaborative LLM Systems routing

- **Wanxing Wu***, He Zhu*, Yixia Li*, Yun Chen, Guanhua Chen† (*Equal contribution)
- *Under Review, 2026* [*arXiv:2602.11877*]

ACADEMIC PROJECTS

ReflexSQL: A Self-Reflective Agent for Robust Text-to-SQL Generation — *Dec 2025 – Jan 2026*
QueryPilots, Institut Polytechnique de Paris [Poster]

- Built an execution-aware self-reflection framework for enterprise-scale Text-to-SQL on Spider 2.0-Lite, combining candidate voting, execution feedback, and memory-augmented exploration.

Conservative Off-Policy Trust Region Policy Optimization *Nov 2025 – Jan 2026*

- Developed a dimension-wise masking strategy for off-policy TRPO that leverages on-policy gradient alignment to suppress conflicting update components, ensuring update directions remain consistent with the on-policy objective; validated on PyBullet/Gym locomotion tasks.

HONORS & AWARDS

- **Outstanding Graduate**, Zhiren College, SUSTech (Jul 2025).
- **Outstanding Graduation Thesis**, Top 5%, SUSTech (Jul 2025).
- Outstanding Student Scholarship, SUSTech (Oct 2024).
- **MCM/ICM Mathematical Contest in Modeling**, Honorable Mention (Feb 2024).

SKILLS

LLM & ML: inference pipeline (vLLM, SGLang), agentic-RL workflows, SFT/RL training, model evaluation.

Programming & Dev: Python, PyTorch, Docker, Linux, Git, shell scripting.

Languages: English (IELTS 6.5, CET-4, CET-6); Chinese (Native).